

INTRODUCTION TO NETWORK PROTOCOL THEORY

1. Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

Steps

1. Open Wireshark.
2. Start capturing on your Wi-Fi/Ethernet interface.
3. Open www.spotify.com.
4. Stop the capture.
5. Use the filter:

dns || tls

Example Packets

Packet No.	Protocol	Info
15	DNS	Standard query A www.spotify.com
16	DNS	Standard query response A 35.186.xxx.xxx
45	TLS (HTTPS)	Client Hello
46	TLS (HTTPS)	Server Hello

2. Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

Command:-

nslookup www.flipkart.com

output:-

Name: flipkart.com

Addresses: 103.243.32.90

3. Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

Feature	HTTP	HTTPS
1. Full Form	HyperText Transfer Protocol	HyperText Transfer Protocol Secure
2. Security	Not secure	Secure
3. Encryption	No encryption	Uses SSL/TLS encryption
4. Default Port	Port 80	Port 443
5. SSL/TLS Certificate	Not required	Required
6. Data Protection	Data can be read by attackers	Data is encrypted and protected
7. Authentication	Does not verify website identity	Verifies website using a digital certificate
8. URL Prefix	http://	https://

**4. Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.

Hint: Look for packets with 'Standard query' and 'Standard query response' in the Info column.**

dns

Example Sequence

Step Packet

- 1 Standard query A www.amazon.in
- 2 Standard query AAAA www.amazon.in
- 3 Standard query response (IPv4 address)
- 4 Standard query response (IPv6 address, if available)

Observation

The browser first sends a DNS query asking for the website's IP address. The DNS server replies with the IP address, and then the browser connects to the website.

5. Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).

Protocol: DHCP (Dynamic Host Configuration Protocol)

Scenario:

When I visit a café and connect my phone to the café's Wi-Fi, the DHCP server in the router automatically assigns my phone an IP address, subnet mask, default gateway, and DNS server. After receiving these network settings, I can open apps like Instagram, YouTube, or WhatsApp without manually configuring the network. DHCP makes connecting to new Wi-Fi networks quick and automatic for all users.